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Unità formativa	Inglese
Argomento	CLOUD - Exploring cloud systems

Instructions

1) You'll be divided in triplets
2) Choose **one** of the topics below:

1. Cloud Computing Fundamentals

- What is Cloud Computing and how does it work?
- The evolution from traditional data centres to the cloud
- Service models: IaaS, PaaS and SaaS
- Public, Private, Hybrid and Multi-Cloud
- Advantages and disadvantages of cloud computing
- Edge Computing vs Cloud Computing
- Serverless Computing
- Cloud-Native Applications

2. Major Cloud Providers

- Comparing AWS, Microsoft Azure and Google Cloud
- Core AWS services
- Core Azure services
- Core Google Cloud services
- Cloud cost management
- Choosing the right cloud provider

3. Virtualization and Containers

- Virtual Machines vs Containers
- How Docker works
- Introduction to Kubernetes
- Container orchestration
- Microservices Architecture
- Infrastructure as Code (IaC)

4. DevOps and Automation

- What is DevOps?
- CI/CD Pipelines
- Git and GitHub for cloud development
- Continuous Integration
- Continuous Deployment
- Monitoring and Logging
- Infrastructure Automation
- Configuration Management

5. Cloud Security

- Shared Responsibility Model
- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Data Encryption
- Backup and Disaster Recovery
- Business Continuity
- Zero Trust Security
- Cloud Security Best Practices
- Cyber threats in the cloud
- GDPR and cloud compliance

6. Cloud Networking

- Virtual Private Cloud (VPC)
- VPN and Hybrid Connectivity
- DNS and Load Balancing
- Cloud Firewalls
- Content Delivery Networks (CDNs)
- Network Security Groups
- Subnets and Routing

7. Storage and Databases

- Object Storage
- Block Storage
- File Storage
- SQL vs NoSQL Databases
- Cloud Database Services
- Backup Strategies
- Data Replication

8. Artificial Intelligence and Cloud

- How cloud computing supports Artificial Intelligence
- Machine Learning as a Service (MLaaS)
- Generative AI in the cloud
- Cloud GPUs
- Big Data Analytics
- Data Lakes
- Cloud for Deep Learning

9. Case Studies

- How Netflix uses cloud computing
- Spotify's cloud infrastructure
- Cloud computing in healthcare
- Cloud solutions for banking
- Smart Cities and the cloud
- Cloud computing in e-commerce
- Cloud and the Internet of Things (IoT)

10. Cloud Operations

- Cloud Migration
- Cloud Monitoring
- Incident Management
- Service Level Agreements (SLAs)
- High Availability
- Fault Tolerance
- Scalability
- Elasticity
- Cloud Cost Optimization
- Green Cloud Computing

11. Emerging Technologies

- Quantum Computing and the Cloud

- Confidential Computing
- FinOps
- Platform Engineering
- Digital Twins
- Edge AI
- Sustainable Cloud Computing

IF YOU ARE ABLE, YOU CAN ALSO OPT FOR:

12. Practical Demonstrations

- Creating a virtual machine in the cloud
- Deploying a website on AWS or Azure
- Creating a Kubernetes cluster
- Deploying a Docker application
- Configuring automated backups
- Monitoring a cloud infrastructure
- Building a CI/CD pipeline
- Configuring a Load Balancer
- Creating a Data Lake
- Managing users and permissions with IAM

3) Prepare a **15/20 minute technical presentation**.

   Your presentation should:

- explain the topic clearly and accurately;
- include technical vocabulary related to your field;
- provide real-world examples or case studies where possible;
- include diagrams, screenshots, or demonstrations if appropriate;
- conclude with a brief summary and a Q&A session.